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Yao Zhang

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1 Recreation

1.1 This Is Us

1.2 Vocabulary

1.3 Conversation

1. 1700 2. 1500 3. 1300

1.4 Driving Experience

座椅调节顺序: 上下 → 前后 → 靠背. 方向盘调节顺序: 前后 → 上下. 后视镜调节顺序: 上下 → 左右.

1. 坐姿后视镜调节
2. 自动挡正确起步和停车
3. 方向盘正确打法
4. 踩刹车技巧
5. 新手开车注意事项
6. 开车安全口诀
7. 提升车感
8. 灯光正确操作
9. 新手开车错误习惯
10. 停车技巧
11. 车头车尾距障碍物距离
12. 倒车入库
13. 开车出库
14. 侧方位停车
15. 窄路侧方位停车
16. 靠边停车
17. 掉头
18. 窄路判断车距
19. 农村开车技巧
20. 市区开车技巧
21. 雨天开车
22. 雾天开车
23. 跑高速流程
24. 跑高速注意事项
25. 雨天跑高速
26. 雾天跑高速
27. 雪后跑高速
28. 夏天开车注意事项
29. 冬天开车正确使用空调
30. 手机导航

1.5 The Big Bang Theory

1.6 Popular Science Series

1. 杨振宁: 美与物理学
2. 杨振宁: 20 世纪数学与物理的分与合
3. 丘成桐: 广义相对论与数学
4. 丘成桐: 时空几何学与广义相对论

5. 高涌泉: 比相对论更奇怪的量子力学
6. 高涌泉: 爱因斯坦如何发现狭义相对论
7. 高涌泉: 20 世纪物理大师理查德 · 费曼其人其事
8. 高涌泉: 为什么非弯曲时空不可? 爱因斯坦的洞见
9. 高涌泉: 量子力学究竟怪在哪里? 海森堡在 100 年前开启的学问
10. 张庆瑞: 量子加 AI, 新文明诞生
11. 张庆瑞: 量子纠缠, 由哲学到数学, 再经科学到科技
12. 李琳山: 信号与人生 2019 2020 2021 2022 2023 2025
13. 张海潮: 爱因斯坦的科学与生平
14. Chad Davies: Maxwell's Equations
15. 邵锦昌: 基础科学研究方法: 跨领域研究实例
16. 陈义裕: 玄妙时空理论的绚丽展现
17. 耿朝强: 宇宙学与粒子物理前瞻问题
18. 沈成平: 从夸克到上帝粒子用粒子物理打开世界真相
19. 余舜德: 为什么人类学需要脑神经科学
20. WSA2025: The Future of Stroke Treatment: Insights from AI and Emerging Technologies
21. M. Poluekto & A. Polar: Efficient training of Kolmogorov-Arnold Networks
22. 时间管理 A, B
23. 宇宙生命科技
24. 访谈: 原生家庭

1.7 The Story of the Stone

1.8 New Concept English Series

1.9 Romance of the Three Kingdoms

1. 宴桃园豪杰三结义, 斩黄巾英雄首立功
2. 张翼德怒鞭督邮, 何国舅谋诛宦竖
3. 议温明董卓叱丁原, 馈金珠李肃说吕布
4. 废汉帝陈留为皇, 谋董贼孟德献刀
5. 发矫诏诸镇应曹公, 破关兵三英战吕布
6. 焚金阙董卓行凶, 匿玉玺孙坚背约

2 Life Sciences

2.1 Cardiology

2.2 Brain Imaging

2.3 Clinical Neurology

2.4 Biomedical Engineering

2.5 Medical Image Computing

2.6 Maternal and Child Health

2.6.1 Job Specifications

1. 岗位认知 A
2. 岗位认知 B
3. 岗位要求 A
4. 岗位要求 B

2.6.2 Infant Care

1. 新生儿的定义及类别 A
2. 新生儿的定义及类别 B
3. 新生儿的生长指标及生理特点 A
4. 新生儿的生长指标及生理特点 B
5. 新生儿的能力发育
6. 婴儿的生理反射和几种状态
7. 婴儿的营养需求
8. 婴儿的喂养方式 A
9. 婴儿的喂养方式 B
10. 婴儿非疾病的多种异常情况 A
11. 婴儿非疾病的多种异常情况 B
12. 婴儿非疾病的多种异常情况 C
13. 婴儿非疾病的多种异常情况 D
14. 婴儿非疾病的多种异常情况 E
15. 满月婴儿的发育特点及指标
16. 婴儿家庭护理准备
17. 婴儿日常护理 A
18. 婴儿日常护理 B
19. 婴儿日常护理 C
20. 婴儿日常护理 D
21. 婴儿日常护理 E
22. 婴儿日常护理 F
23. 婴儿日常护理 G
24. 婴儿日常护理 H

25. 婴儿日常护理 I
26. 新生儿常见问题及紧急情况处理 A
27. 新生儿常见问题及紧急情况处理 B
28. 婴儿早教和智力开发 A
29. 婴儿早教和智力开发 B
30. 婴儿早教和智力开发 C
31. 婴儿日常护理: 游泳
32. 婴儿日常护理: 疾病的预防与观察
33. 婴儿常见疾病的观察和护理 A
34. 婴儿常见疾病的观察和护理 B
35. 婴儿常见疾病的观察和护理 C
36. 婴儿常见疾病的观察和护理 D
37. 婴儿常见疾病的观察和护理 E
38. 婴儿常见疾病的观察和护理 F
39. 婴儿出行安全
40. 早产儿
41. 先天性异常儿 A
42. 先天性异常儿 B
43. 婴儿护理实践操作

2.6.3 Maternal Care

1. 产褥期产妇身体变化特点 A
2. 产褥期产妇身体变化特点 B
3. 产褥期的临床表现 A
4. 产褥期的临床表现 B
5. 产褥期护理的重要性 A
6. 产褥期护理的重要性 B
7. 产褥期护理应注意的问题
8. 不同种类的产褥期护理
9. 产褥期的营养调配 A
10. 产褥期的营养调配 B
11. 产后恢复及产后调养 A
12. 产后恢复及产后调养 B
13. 母乳喂养 A
14. 母乳喂养 B
15. 母乳喂养 C
16. 母乳喂养的好处
17. 哺乳指导 A
18. 哺乳指导 B
19. 母乳喂养的常见问题 A
20. 母乳喂养的常见问题 B
21. 母乳喂养的常见问题 C
22. 产妇护理实践操作

2.7 Fundamental Statistical Genomics

2.8 Machine Learning for Neuropsychiatric Disorders

3 Computer Science

3.1 Deep Learning

1. (a) Introduction
   (b) Overview of CV on Image Classification
2. Neural Network
3. CNN
4. (a) Training a Neural Network
   (b) Better Accuracy
   (c) Optimizer
   (d) Regularization
   (e) Nuts and Bots of Deep Learning
5. Advanced CNN Architecture
6. (a) RNN
   (b) Transformer
   (c) Vision Transformer
7. (a) Pruning
   (b) Quantization
   (c) Tensor decomposition
   (d) Low complexity model
8. (a) Semantic Segmentation
   (b) Object Detection
   (c) Object Detection New Trends
9. (a) GAN
   (b) Diffusion Model
10. (a) Pretrained Model and LLM
    (b) Self Supervised Learning

3.2 Causal Data Science

1. Introduction
2. Markov Equivalence Class and D-Separation
3. Markov Equivalence Class and Ancestral Graphs
4. Partial Ancestral Graph and Fast Causal Inference Algorithm
5. Conditional Independence Tests & Possible D-Separating Sets
6. Midterm Review
7. Effect Identification
8. Adjustment Criterion for PAGs
9. Do Calculus
10. ID Algorithm
11. ID Algorithm for PAGs

12. Counterfactuals and Effect Estimation

3.3 Mathematics of Deep Learning

1. The Curse of Dimensionality in Machine Learning
2. Function Approximation. Linear vs Non-Linear Approximation
3. Universal Approximation Theorem for Shallow Nets. Harmonic Analysis for Shallow Nets
4. Reproducing Kernel Hilbert Spaces
5. Learning in the Overparametrised Regime. Generalisation Bounds for Normed spaces
6. Convex Optimization Reminders
7. Optimization in Shallow Spaces: the linear regime. The Neural Tangent Kernel
8. Nonlinear overparametrised regime. Mean-Field description
9. Lower bounds: Statistical-Queries, Reductions
10. Depth Separation
11. Invariances and Symmetries
12. Implicit Regularisation. Noisy dynamics. Gradient Flow vs Gradient Descent vs Stochastic Gradient Descent
13. Generative Modeling: Diffusion

3.4 Generative Artificial Intelligence

1. Introduction to Machine Learning
2. Back Propagation and Training Process
3. Back Propagation and Cost Function
4. Loss Function of Generative AI
5. Generative AI Architecture & Entropy Loss Function
6. Overall Perspectives of Generative AI Architecture
7. Sequential generative AI with RNN
8. Back Propagations in RNN

9. Attention is All You Need
10. Multi-Head Attentions in Transformer Encoder
11. Cross-Attentions in Transformer Encoder-Decoder
12. Generative AI for Image, Generative Adversarial Network Architecture & Loss Functions
13. Variational Autoencoder Architecture
14. Variational Autoencoder Loss Function and Image Generation
15. Diffusion Model for Image Generation
16. Reinforcement Learning for Fine Tuning and Human Feedback
17. Markov Property and Bellman Equation
18. Monte Carlo Method & Temporal Difference Method
19. Deep Reinforcement Learning for Value Agent Network
20. Deep Reinforcement Learning for Policy Agent Network

3.5 Foundations of Machine Learning

3.6 Nonconvex Optimization for Deep Learning

0. 00
1. 01
2. 02
3. 03
4. 04
5. 05
6. 06
7. 07
8. 08
9. 09
10. 10
11. 11

3.7 Machine Learning and Reinforcement Learning

3.8 Artificial Intelligence in the Sciences and Engineering

3.8.1 2024

1. Course Introduction
2. Introduction to Deep Learning A
3. Introduction to Deep Learning B
4. Importance of PDEs in Science
5. PINNs: Introduction
6. PINNs: Limitations and Extensions A
7. PINNs: Limitations and Extensions B
8. PINNs: Theory A
9. PINNs: Theory B
10. Introduction to Operator Learning A
11. Introduction to Operator Learning B
12. Fourier Neural Operators

13. Spectral Neural Operators and Deep Operator Networks
14. Convolutional Neural Operators
15. Time-Dependent Neural Operators
16. Large-Scale Neural Operators
17. Attention as a Neural Operator
18. Windowed Attention and Scaling Laws
19. Introduction to Hybrid Workflows A
20. Introduction to Hybrid Workflows B
21. Neural Differential Equations NODE
22. Introduction to Diffusion Models
23. Introduction to JAX
24. Symbolic Regression and Model Discovery
25. Applications of AI in Chemistry and Biology A
26. Applications of AI in Chemistry and Biology B
27. Materials

3.8.2 2025

1. Course Introduction
2. Introduction to Deep Learning
3. Physics-Informed Neural Networks
   (a) Introduction
   (b) Theoretical Insights
4. Operator Learning
   (a) Introduction
   (b) FNO
   (c) ReNO
   (d) Operator Transformer
   (e) Graph-based Model
   (f) GAOT
5. Generative model for PDE: GenCFD
6. Foundation models of PDEs: Poseidont
7. Applications of AI in Chemistry Biology: A B

3.8.3 Kolmogorov-Arnold Networks

1. Kolmogorov-Arnold Networks
2. Efficient training of Kolmogorov-Arnold Networks
3. Sparsity and Symbols with Kolmogorov-Arnold Networks
4. A Revisit of Kolmogorov's mapping theorem for Neural Network

3.8.4 Physics Informed Machine Learning

1. PIML in AISE
2. AI/ML+Physics
   (a) Choosing what to model
   (b) Curating Training Data
   (c) Designing an Architecture
   (d) Crafting a Loss Function

(e) Employing an Optimization Algorithm

(f) Recap and Summary

(g) Preview of Upcoming

3. Autoencoders & PIML

4. Sparse Identification

(a) Sparse Machine Learning Models

(b) Training Data & Disambiguating Models

(c) Effective Coordinates

(d) The Library of Candidate Nonlinearities

(e) The Optimization Algorithms

5. Discrepancy Modeling with PIML

6. Hamiltonian Neural Networks

7. Lagrangian Neural Network

8. Neural Implicit Flow

9. Neural ODEs

10. Python Symbolic Regression

11. Residual Networks

12. Fourier Neural Operator

13. Deep Operator Networks

14. Physics Informed Neural Networks

3.8.5 Transformers from an Optimization Perspective

3.9 Biologically Inspired Artificial Intelligence and Optimization

1. (a) Introduction to the Course & Its Policies & Its Motivation

   (b) Evolutionary Approach to Engineering Design Optimization

   (c) Population Genetics of Evolutionary Algorithms

   (d) The Four Forces of Evolution and The Drift Barrier

   (e) The Basic Genetic Algorithm and Its Implementation

   (f) GA Wrap: Options for Selection & Crossover & Mutation & Extensions

2. (a) Evolutionary Computing from Optimization to Programming

   (b) Genetic Programming, Immunocomputing & Artificial Immune Systems

   (c) Immunocomputing: Genetic Approaches for Diverse Solution Portfolio

3. (a) Multi-Criteria Decision Making, Pareto Optimality & Intro to MOEA

   (b) Multi-Objective Genetic Algorithms: Weight/Vector-Based Approaches

   (c) Pareto Ranking and Moving from Communities to Meta-Populations

4. (a) From DGA/PGA to Niching Methods for Multi-Modal Optimization

   (b) Niching Methods in Multi-Modal Optimization

5. (a) Introduction to Simulated Annealing and Entropy

(b) From Maximum Entropy Toward Optimization by Sim Annealing

(c) Toward SA: Intro to Boltzmann Sampling and Monte Carlo Integration

(d) Simulated Anneal & Wrap and Introduction to Ant Colony Opt

6. (a) Ant Colony Opt. and Intro. to Bacterial Foraging Opt.

   (b) Bacterial Foraging Opt. & Particle Swarm Optimization

7. (a) Neural Foundations of Learning

   (b) Feeding Forward from Neurons to Networks

   (c) Recurrent Networks and Temporal Supervision

   (d) Reinforcement Learning: Active Learning in Rewarding Environments

   (e) Learning without Teacher: Unsupervised & Self-Supervised Learning

   (f) Spiking Neural Networks and Neuromorphic Computation

8. (a) Complex Systems Models of Computation: Cell. Automata & Neighbors

3.10 High Dimensional Analysis: Random Matrices and Machine Learning

1. Introduction and Survey
2. Volumes in High Dimensions
3. Concentration of Volumes
4. Gaussian Random Vectors and Concentration of Their Norm
5. Exponential Concentration of Norm of Gaussian Random Vectors
6. Non-Linear Concentration of Gaussian Random Vectors for Lipschitz Functions
7. Proof of Non-Linear Concentration for Gaussian Random Vectors
8. General Remarks on Linear and Non-Linear Concentration Inequalities
9. Wishart Random Matrices and Concentration of Largest Eigenvalue
10. Proof of Concentration of Largest Eigenvalue
11. The Marchenko-Pastur Law for Wishart Matrices
12. Preparations for Proof of Marchenko-Pastur Law
13. Proof of Marchenko-Pastur: Equation for Stieltjes Transform

14. Proof of Marchenko-Pastur: Stieltjes Inversion Formula
15. Spiked Signal-Plus-Noise Model
16. Proof of the Signal-Plus-Noise Theorem
17. Double Descent and Linear Regression: Over-Determined Case
18. Double Descent and Linear Regression: Under-Determined Case
19. General Remarks on Random Feature Model
20. Asymptotic Eigenvalue Distribution in the Random Feature Model
21. Another Proof of Marchenko-Pastur – Via Stein's Identity
22. Resolvent Method and Cumulant Expansion
23. Cumulants and Their Properties and Uses
24. Calculation of the Random Feature Eigenvalue Distribution
25. Gaussian Equivalence Principle for Non-Linear Random Features
26. Gradient Descent for Linear Regression
27. Time Evolution of Learning and Neural Tangent Kernel
28. Properties of the Neural Tangent Kernel
29. Free Probability Theory and Linearization of Non-Linear Problems
30. The curse and the blessing of high dimensions

3.11 Mathematics of Massive Data Analysis: Fundamentals and Applications

1. Concentration Inequalitie; Geometry of High-Dimensional Euclidean Space
2. Hyperspheres and Hyperballs are Weird in High Dimensions
3. Gaussian Annulus Theorem and Johnson-Lindenstrauss Lemma
4. Markov Chains and MCMC
5. Convergence of Random Walks on Undirected Graphs Using Conductance
6. More Markov Chains
7. (a) Volume via MCMC and PageRank

(b) Overview

8. Overview of Hashing, Sketching, and Random Graph Theory

9. (a) Overview of Wavelets, Low-Dimensional Embeddings, and Persistent Homology

   (b) Hashing Theory

10. K-Independent Hash Functions

11. Frequency Moments of Data Streams and an Idealized Count-Distinct Sketch

12. Non-Idealized Count-Distinct Sketch and F2 Sketch

13. Frequent Items Sketch, MinHash Sketch

14. Erdos-Renyi Random Graphs and Phase Transitions

15. Erdos-Renyi Diameter and Isolated Vertices Phase Transitions

16. Erdos-Renyi Component Size Phase Transitions

17. Percolation Theory

18. Finishing up Percolation Theory, and Intro to Wavelets

19. Finishing up Haar Wavelet

20. Necessary Conditions for General Systems of Wavelets

21. Sufficient Conditions for General Systems of Wavelets. The proof of lemma 11.6 is incorrect and revised as computation 11.6.

22. The Fast Fourier Transform. Symmetry if real proof error; complex conjugates added.

23. Matrix Sketching

24. VC-Dimension

25. Generalization Error in Classification

26. Representing Topological Spaces by Decomposing into Complexes

27. More Complexes and Introduction to Chains

28. Computing Simplicial Homology

29. Morse Functions

30. Morse-Smale Complex and Piece-Wise Linear Morse Functions

31. HyperMinHash

32. Secure Federated Count-Queries

4 Mathematical Foundations

4.1 Analysis

Analysis I

1. Real Numbers and the completeness

2. Limits and continuity in metric space

3. Open and closed sets in metric space

4. Topological space & The closure of the set

5. Equivalence of continuous functions & Inheritance Principle & Homeomorphism

6. Product metric spaces & Completeness & Compactness (sequentially compact)

7. Compactness (2) & Bolzano-Weierstrass Thm & Nested seq of nonempty compact sets

8. ontinuous functions and Homeomorphisms on compact sets & Connectedness

9. Some properties in topological spaces

10. Accumulation points and isolated points & Dense & Hausdorff space

11. Properties of Connectedness

12. Path connected & Compactness in metric space, Hausdorff space, Topological space(1)

13. Compactness (2) & Sequentially compact implies compact(open cover) (1)

14. Sequentially compact implies compact(open cover) (2) & Totally bounded (1)

15. Totally bounded (2) & Perfect & Cantor set

16. Cantor surjection Theorem

17. Peano Curves & Riemann integrable & Darboux integrable(1)

18. Darboux integrable(2) & Riemann integrable iff Darboux integrable(1)

19. Riemann integrable iff Darboux integrable(2) & Some integrable functions (1)
20. Some integrable functions (2) & Lebesgue number & Zero set
21. Oscillation & Riemann-Lebesgue Theorem
22. Some applications of R-L theorem
23. Fundamental Theorem of Calculus & Convergence in functional space
24. Complete functional space & Convergence function series & Integrals wst unif conv.
25. Derivatives w.s.t uniform convergence & Equicontinuity and Arzela-Ascoli Theorem
26. Heine-Borel Theorem in a function space & Stone-Weierstrass Theorem
27. Fixed point, contraction and ODE
28. Nowhere differentiable continuous function & Baire' s Theorem
29. Corollary of Baire' s Theorem
30. Multiple integral and Fubini' s Theorem
31. Change of variables formula (1)
32. Change of variables formula (2)
33. Banach spaces
34. Banach algebra
35. Differentiation in Banach Spaces
36. Some examples & Integration in Banach spaces (1)
37. Integration in Banach spaces (2)
38. Inverse function theorem
39. Lebesgue Outer measure
40. Lebesgue measurable set & Abstract outer measure
41. Measurable Sets form Sigma-Algebra & Measure continuity
42. Regularity (1)
43. Regularity (2)

44. Every open set is a countable disjoint union of balls plus a zero set

45. Measure Product & Inner measure (1)

46. Inner measure (2) & Slice Measure (1)

47. Slice Measure (2) & Lebesgue integrable functions & Monotone Convergence Theorem

48. Completed undergraph & Dominated Convergence Thm and Fatou' s Lem & Basic properties

49. Some basic properties of measurable functions (2)

50. Cavalieri's principle & Equivalence measurable function definition & Tonelli Thm

51. Vitali covering Lemma

52. Density Theorem

Analysis II

1. Lebesgue Differentiation Theorem (1)

2. Lebesgue Differentiation Theorem (2) & Absolutely continuous functions

3. Indefinite integral is abs conti. & A monotone function has a derivative a.e. (1)

4. A monotone function has a derivative almost everywhere (2)

5. Riemann integrals vs Lebesgue integrals

6. Bounded variation

7. Equi relation of B-V functions and abs conti functions & R-S integrals (1)

8. Riemann-Stieltjes integral (2)

9. Riemann-Stieltjes integral (3)

10. Riemann-Stieltjes integral (4)

11. L^2 $[a, b]$ space

12. Properties of L^2 $[a, b]$ space

13. Orthogonal basis in L^2

14. Fourier coefficients and Parseval's formula

15. A necessary condition of Fourier series converges in L^1 & Riemann-Lebesgue lemma

16. Dirichlet kernel

17. Dini Theorem & There is a continuous function whose Fourier series diverges (1)

18. There is a continuous function whose Fourier series diverges (2) & U-B Principle

19. There is a continuous function whose Fourier series diverges (3)

20. Some properties of Fourier coefficients & Cesàro summability

21. Fejér's Theorem & Approximate integrable functions by tri-polynomial in L^1 norm

22. The average of Fourier series converges in L^p norm

23. Equi. of the average of Fourier series converges unif & Weakly convergence(1)

24. Weakly convergent(2) & Equi condition for tri-series to be the Fourier series

25. Convergence of Cesàro mean of Fourier series (1)

26. Part2 Conv of Cesàro mean of Fourier series(2), Conjugate Fourier series & Dini theorem

27. Singular integrals and conjugate function

28. Bound of the L^2 norm of conjugate function & Hardy-Littlewood Maximal function

29. Bound of the norm of the supremum of conjugate functions

30. A counterexample & C-Z Decomposition & The conjugate function exists a.e. in L^1

31. The conjugate function exists a.e. in L^1 (2)

32. The conjugate function exists a.e. in L^1 (3) & Weakly estimate

33. Applications of Fourier series -isoperimetric inequality

34. Applications of Fourier series -continuous but nowhere differentiable function

35. Fourier transform and some properties

36. Fourier inversion formula and Minkowski's inequality

37. Poisson summation formula & Roth's theorem on 3-term AP

38. Discrete Fourier series, Fourier transform and some properties (1)

39. Discrete Fourier series, Fourier transform and some properties (2)

40. Bourgain's theorem

41. Proof of Roth's theorem on 3-terms AP (step 1)
42. Proof of Roth's theorem on 3-terms AP (step 2 to step 7)
43. Proof of Roth's theorem on 3-terms AP (step 9 , step 8)
44. Proof of Roth's theorem on 3-terms AP (step 8)
45. There is f in L^1 such that for a.e. x, the Fourier series diverges (1)
46. There is f in L^1 such that for a.e. x, the Fourier series diverges (2)
47. There is f in L^1 such that for a.e. x, the Fourier series diverges (3)
48. There is f in L^1 such that for a.e. x, the Fourier series diverges (4)

4.2 Calculus

Calculus I

1. System of real number 1
2. System of real number 2
3. System of real number 3
4. Completeness of Real Number, Weierstrass Theorem 1
5. Weierstrass Theorem 2, define subsequence, Cauchy Sequence
6. Least Upper Bound & Greatest Lower Bound
7. Continuous function, uniformly continuous, Lipchitz continuous
8. Intermediate value theorem, Extreme Value Theory
9. Rieman sum
10. Properties of integral, MVT/general MVT for integral, logarithm
11. Logarithm and exponential 1
12. Logarithm and exponential 2
13. Fundamental Theorem of Calculus 1, Rolle's Theorem, MVT for derivative 1
14. MVT for derivative 2, approximation by linear function, FTC 2
15. Properties of derivative, composite functions, chain rule

16. First/second derivative test
17. Order of magnitude, introduce an smooth function
18. Some example of oscillating functions, chain rule, method of substitution
19. Integration by parts
20. Integration of rational functions 1
21. Integration of rational functions 2, Improper integral 1
22. Improper integral 2
23. Taylor's series: Gamma function, power series, expansion of the logarithm
24. Taylor's series: Taylor's theorem, Cauchy's & Lagrange's remainder
25. Taylor's series: estimate remainder term, expansion of elementary
26. Taylor's series: examples
27. Taylor's polynomial: interpolation 1
28. Taylor's polynomial: interpolation 2, approximation in first order
29. Taylor's polynomial: approximation in second order
30. Taylor's polynomial: fixed point approximation, Stirling's formula
31. Series: concepts of convergence and divergence 1
32. Series: rearrangement, tests for absolute convergence and divergence 1
33. Series: tests for abs. convergence and divergence 2, sequences of functions
34. Series: pointwise/uniform convergence 1
35. Series: pointwise/uniform convergence 2
36. Series: power series, interval of convergence
37. Series: product of two power series
38. Series: expansion of given power series, infinite product
39. Fourier series: periodic function, complex notation 1
40. Fourier series: complex form 2, trigonometric formula, Riemann Lebesgue Lemma

41. Fourier series: examples of Fourier expansion
42. Fourier series: main theorem on Fourier expansion
43. Fourier series: examples of Fourier series 1
44. Fourier series: examples of Fourier series 2, Bessel's Inequality
45. Approximation by trigonometric and rational polynomial 1
46. Approximation by trigonometric and rational polynomial 2
47. Approximation by trigonometric and rational polynomial 3
48. Inner product
49. Bernoulli polynomial and their applications 1
50. Bernoulli polynomial and their applications 2

Calculus II

1. Functions of multiple variables & partial derivative
2. Continuity
3. Differentiability & directional derivative (1)
4. Directional derivative (2) & tangent plane
5. Change of variables & Taylor series
6. MVT & Taylor expansion & Integral
7. Double integral & length of curve
8. Curvature & linear differential one form
9. Line integral
10. Heine-Borel theorem
11. Compact subset & Implicit Function theorem (1)
12. Implicit Function theorem (2)
13. Inverse Function theorem (1)
14. Inverse Function theorem (2) & extreme value

15. Lagrange Multiplier method
16. Examples & Hölder Inequality
17. sufficient condition for local max/min
18. Jordan measurable
19. Jordan area (1)
20. Jordan area (2)
21. Double integral (1)
22. Double integral (2)
23. Transformation of multiple integrals
24. Improper multiple integrals (1)
25. Improper multiple integrals (2) & volumes
26. Surface area
27. Surface area formula
28. Multiple integral in curved coordinate
29. Extend to $\mathbb{R}^n$
30. Integral of unbounded set
31. Fourier integral theorem (1)
32. Fourier integral theorem (2)
33. Fourier transform - decay property & Parseval's identity
34. Fourier transform of several variables
35. Green theorem
36. Divergence theorem
37. Prove Inverse Function theorem by Green theorem
38. Orientation of surface in $\mathbb{R}^3$ I
39. Orientation of surface in $\mathbb{R}^3$ II

40. Gauss theorem in $\mathbb{R}^3$ I
41. Gauss theorem in $\mathbb{R}^3$ II
42. Application of fluid
43. Stoke's theorem
44. General surface (partition of unit) (1)
45. General surface (partition of unit) (2)
46. General surface (partition of unit) (3) & functions of one complex variable
47. Complex function - power series
48. Complex function - differentiability
49. Conformal map & integration of analytic functions - Cauchy's theorem
50. Complex integral theorem & Cauchy integral formula
51. Zeros, poles and residues

Orthogonal Curvilinear Coordinates

4.3 Lie Algebra

1. Introduction to Lie Groups with Focus on Matrix Groups A
2. Introduction to Lie Groups with Focus on Matrix Groups B
3. Introduction to Lie Groups with Focus on Matrix Groups C
4. Definition, Examples, Lie Subalgebra, Ideals, Adjoint Homomorphism
5. Adjoint Homomorphism, Structure Constants, Cross Product Lie Algebra
6. Ideals, Low Dimensional Lie Algebras, Derived Algebra, Solvable, Radical A
7. Ideals, Low Dimensional Lie Algebras, Derived Algebra, Solvable, Radical B
8. Ideals, Low Dimensional Lie Algebras, Derived Algebra, Solvable, Radical C
9. Solvable, Nilpotent and Simple Lie Algebras A
10. Solvable, Nilpotent and Simple Lie Algebras B
11. Lemmas, Engel and Lie's Theorems, Lie Algebra Homomorphism, Modules A

12. Lemmas, Engel and Lie's Theorems, Lie Algebra Homomorphism, Modules B
13. Representations, Factor Modules, Irreducible Submodules A
14. Representations, Factor Modules, Irreducible Submodules B
15. Schur's Lemma, SL(2,C) Representation on Vd Begins
16. SL(2,C) Representation on Vd Continued, Irreducibility Confirmed

4.4 Linear Algebra

Linear Algebra I

1. Fields, Vector Spaces, Subspaces.
2. Subspaces, Linear Dependence, Basis.
3. Linear Dependence, Basis.
4. Basis, Replacement Theorem, Dimension.
5. Linear Transformation, Kernel, Range, Dimension Theorem.
6. Dimension Theorem, Projection.
7. Matrix Representations.
8. Matrix Representations, Invertible Linear Transformations.
9. Invertible Linear Transformations, Change of Coordinates.
10. Elementary Matrices.
11. Elementary Matrices, Gaussian Elimination.
12. Gaussian Elimination, Determinant of Order 2.
13. Determinants.
14. Determinants(in Terms of Permutations),Cramer's Rule,Adjoint Matrices.
15. Diagonalization
16. Diagonalization
17. Invariant Subspaces, Cayley-Hamilton Theorem
18. Jordan forms, generalized eigenspaces (w. notes)

19. Jordan forms
20. Jordan forms.
21. Jordan forms.
22. Exponential of matrices
23. Systems of first order differential equations, minimal polynomials.
24. Rational canonical forms.
25. Rational canonical forms.
26. Rational canonical forms.

Linear Algebra II

1. Dual Spaces
2. Quotient Spaces
3. Inner Product Space
4. Inner Product Space, Gram-Schmidt Process
5. Orthogonal Projection
6. Hilbert Space, Riesz Representation Theorem
7. Riesz Representation Theorem
8. Adjoint Operators, Normal Operators
9. Normal and Self adjoint Operators
10. Positive semidefinite, Orthogonal and Unitary operators
11. Orthogonal and Unitary operators
12. Orthogonal projection, Spectral theorem
13. Spectral theorem,Simultaneous diagonalization of normal operators
14. Singular value decomposition, Polar decomposition
15. Pseudoinverse
16. Pairs of commuting matrices over a finite field

17. Pairs of commuting matrices over a finite field
18. Bilinear forms
19. Bilinear forms, Quadratic forms
20. Quadratic forms
21. Quadratic spaces
22. Quadratic spaces
23. Witt decomposition theorem
24. Cartan-Dieudonné theorem
25. Cartan-Dieudonné theorem
26. Bézout's theorem
27. Bézout's theorem

4.5 Basic Geometry

4.6 Measure Theory

1. Introduction: a Non-Measurable Set
2. Classes of Subsets (Semi-Algebras, Algebras and Sigma-Algebras), and Set Functions
3. Set Functions
4. Caratheodory Theorem
5. Monotone Classes
6. The Lebesgue Measure I
7. The Lebesgue Measure II
8. Complete Measures
9. Approximation Theorems
10. Integration: Measurable and Simple Functions
11. Measurable Functions

12. Definition of the Integral

13. Integral of Simple Functions

14. Properties of the Integral I

15. Properties of the Integral II

16. Theorems on the Convergence of Integrals

17. Product Measures

18. Measure on a Countable Product of Spaces

19. Fubini's Theorem

20. Hahn-Jordan Theorem

21. Radon-Nikodym Theorem

22. Almost Sure and Almost Uniform

23. Convergence in Measure

24. Hölder and Minkowski Inequalities

25. L_p Spaces

26. From Convergence in Measure to Convergence in L_p

27. Bounded Linear Operators in L_p

28. Vitali's Covering Lemma

29. Differentiability of Functions of Bounded Variations

30. Absolutely Continuous Functions

31. Decomposition of Distribution

32. Cantor Ternary Set and Function

33. Solution of Exercise 3.2.4

34. Solution of Exercise 6.4.3

35. Solution of Exercise 9.2.3

36. Materials

4.7 Abstract Algebra

Abstract Algebra I

1. Basic notion, Symmetry groups
2. Cosets, Normal Subgroups, Quotient Groups, Isomorphism Theorem
3. Isomorphism Theorem, Tower of Subgroups
4. Butterfly lemma, Jordan-Hölder theorem
5. Jordan-Hölder theorem, Cyclic groups
6. Group Action, Simpleness of A_n
7. Simpleness of A n, Sylow theorems
8. Sylow theorems
9. Semidirect products
10. Structure theorem of finite abelian groups
11. Ring Isomorphism Theorem
12. Chinese Remainder Theorem, Localization
13. ED, PID, UFD
14. Gauss lemma, Criteria for irreducibility
15. Algebraic extensions
16. Algebraic closures
17. Algebraic closures, Normal extensions
18. Separable extensions
19. Primitive element theorem, Finite fields
20. Galois theory
21. Galois theory
22. Galois theory, roots of unity, cyclotomic extensions.
23. Norms and traces.

24. Purely inseparable extensions, cyclic extensions.

25. Solvable extensions, solvable by radicals.

Abstract Algebra II

1. Module theory.

2. Free modules, projective modules.

3. Modules over PID.

4. Modules over PID.

5. Tensor products.

6. Modules of fractions.

7. Noetherian modules.

8. Primary decompositions.

9. Primary decompositions, Nakayama's lemma.

10. Nakayama's lemma, filtered and graded modules, Artin-Rees.

11. Hilbert polynomial, Artinian modules.

12. Abelian categories, general homology theory.

13. Injective modules.

14. Homotopies of morphisms of complexes, derived functors.

15. Ext functor.

16. Ext functor, Ext^1, Tor functor.

17. Group cohomology.

18. Group cohomology, Semisimple modules.

19. Semisimple modules, Semisimple rings.

20. Density theorem, Wedderburn theorem, Burnside theorem.

21. Representations, Characters.

22. Characters, Class functions.

23. Orthogonality relations.

24. Orthogonality relations.

25. Induced characters, Induced representations.

26. Induced representations, Mackey's criterion.

27. Artin theorem, Brauer theorem.

4.8 Complex Analysis

1. 0920 ntroduction to the course, examples of applications, definition of holomorphic functions, algebraic stability properties of holomorphic functions.

2. 0921 Convergent power series are holomorphic. Examples and counterexample (the complex conjugate function).

3. 0927 Holomorphy and differentiability; the Cauchy-Riemann equations. Line integrals.

4. 0928 Line integrals and primitives.

5. 1004 Chapter 3: Cauchy's Theorem. Goursat's Theorem, existence of primitives in a circle, Cauchy's Integral Formula.

6. 1005 Chapter 3: proof of Goursat's Theorem.

7. 1011 Chapter 4: applications of Cauchy's Theorem and integral formula: analyticity, Cauchy's inequalities for derivaties, Liouville's Theorem.

8. 1012 Chapter 4: zeros of holomorphic functions, analytic continuation.

9. 1018 Chapter 4: proof of the principle of analytic continuation. Limits of holomorphic functions, Morera's theorem.

10. 1019 Chapter 4: holomorphic functions defined by integrals

11. 1025 Chapter 5: singularities and meromorphic functions, residue theorem.

12. 1026 Chapter 5: residue theorem and examples.

13. 1101 Chapter 5: meromorphic functions, counting zeros, open image and maximum modulus principle.

14. 1102 Chapter 5: meromorphic functions, counting zeros, open image and maximum modulus principle.

15. 1115 Chapter 6: Eta, THeta, Zeta (a long example). Definitions of the functions, infinite products..
16. 1116 Chapter 6: Eta, THeta, Zeta. Analytic continuation of the zeta function, application to prime numbers.
17. 1121 Chapter 6: Eta, THeta, Zeta (a long example). Sketch of Riemann's approach to counting primes; the Riemann Hypothesis.
18. 1123 Chapter 7: Homotopy and applications. Definition and statement of Cauchy's Theorem for homotopic curves.
19. 1129 Chapter 7: Proof of Cauchy's Theorem for homotopic curves.
20. 1130 Chapter 7: simply-connected open sets, existence of primitives. The complex logarithm.
21. 1206 Chapter 7: The residue theorem and homotopy; winding numbers.
22. 1207 Chapter 8: conformal mapping (definition, first examples).
23. 1213 Chapter 8 (conformal mapping): more examples, statement of Riemann's mapping theorem. Outline of the proof. Schwarz Lemma, automorphisms of the disc.
24. 1214 Chapter 8 (conformal mapping); reduction of Riemann's Theorem to the existence of an extremum.
25. 1220 Chapter 8 (conformal mapping): end of the proof of Riemann's Theorem; Montel's Theorem. Final remarks.
26. 1221 Review of the course, questions

4.9 Matrix Lie Groups

1. Complex and Quaternions A
2. Complex and Quaternions B
3. More Quarternions, Group Theory A
4. More Quarternions, Group Theory B
5. Chapter 2 and Isometries over R,C and H A
6. Chapter 2 and Isometries over R,C and H B
7. Path Connectedness for SO(n) and SU(2) A
8. Path Connectedness for SO(n) and SU(2) B
9. Lie Groups and Tangent Space at I A
10. Lie Groups and Tangent Space at I B
11. What is the Logarithm A
12. What is the Logarithm B

13. On the Derivative of a Homomorphism
14. Adjoint and Lie Algebra A
15. Adjoint and Lie Algebra B
16. Lie Algebra Example 2nd Order BCH A
17. Lie Algebra Example 2nd Order BCH B
18. BCH identity A
19. BCH identity B
20. BCH calculation A
21. BCH calculation B
22. Universal Covers and Lie's Theorem A
23. Universal Covers and Lie's Theorem B
24. Group Orbits and Homogeneous Space A
25. Group Orbits and Homogeneous Space B

4.10 Functional Analysis

1. Linear Spaces: Definition, Examples and Linear Span.
2. Linear Spaces: Quotient Spaces and Convex Sets.
3. Normed Linear Spaces: Definition and Basic Properties.
4. Completing a Normed Linear Space.
5. Finite Dimensional Linear Spaces.
6. Examples of Normed Linear Spaces.
7. In Infinite Dimensions the Unit Ball is not Compact.
8. Zorn's Lemma.
9. The Hahn-Banach Theorem.
10. Convex Sets and Gauge Functions.
11. Geometric Hanh-Banach Theorems.
12. Dual o a Normed Linear Space.
13. Extension of Bounded Linear Functionals, Closed Linear Spans.
14. Reflexive Spaces.
    (a) The Dual Space of C([a,b]).
    (b) An Application of the Hahn-Banach Theorem: the Moment Problem and Chebyshev Approximation.
    (c) A Dual Variational Problem in Optimal Control.

(d) An Application of the Hahn-Banach Theorem: the Existence of a Green Function.

15. Hilbert Spaces.

16. Closed Convex Subsets of a Hilbert Space.

17. Riesz and Lax-Milgram Representation Theorems.

18. Orthonormal Sets and Closed Linear Spans.

19. Orthonormal Bases.

    (a) A Quadratic Variational Problem.

    (b) The Dirichlet Principle.

    (c) Generalized Derivatives and Sobolev Spaces.

20. Uniform Boundedness Principle.

21. Weak Convergence.

22. Uniform Boundedness of Weak Converging Sequences.

23. Weak Sequentially Compactness.

24. Weak* Topology.

25. Applications of Weak Convergence.

26. Bounded Linear Operators.

27. Transpose of Bounded Linear Operators.

28. Strong and Weak Convergence of Operators.

29. Principle of Uniform Boundedness for Maps and Compositions.

30. Open Map Principle.

31. The Closed Graph Theorem.

32. Examples of Bounded Linear Maps: Integral Operators.

33. Symmetric Operators.

34. Eigenvalues of Compact Symmetric Operators.

35. The Fredholm Alternative.

36. An Application to Integral Operators.

37. Materials

4.11 Mathematical Modeling

4.12 Partial Differential Equations

1. A. Intro, definitions, elementary examples.

   B. Advection equations. Characteristic lines. General solutions of quasi-linear first order equations.

2. First order quasi-linear PDEs in two variables. Solution surface. Characteristic equations. Cauchy problem. Examples: advection equations and Burgers equation.

3. Burgers equation. Gradient catastrophe, weak solutions, shock formation, properties and propagation.

4. A. Shock waves, weak solutions, speed of propagation in Burgers equation. Conservation laws.

   B. Parametric solution to general first order PDEs. Monge cone. Characteristic equations.

5. Cauchy problem for general first order PDEs. Wave fronts on an ellipse, caustics. ODEs general solution for a linear system.

6. Second order PDEs intro. Wave equation D'Alembert solution. Changes of variables. Characteristic coordinates.

7. Linear and quasi-linear second-order equations. Characteristic curves.

8. Linear second-order equations. Example: Tricomi equations. Elliptic equations.

9. Parabolic linear second order equations. Examples. Rescaling the dependent variable to remove first order terms.

10. A. Elliptic equations. Laplace equation solutions and Cauchy's theorem. On a disk.

    B. Elliptic equations. Laplace equation solutions and Cauchy's theorem. On the upper half plane.

11. Uniqueness theorems, Laplace equation, divergence theorem, separation of variables, mean value theorem.

12. Maximum minimum principle for elliptic equations. Green's functions. Symmetrical solutions, boundary conditions.

13. Symmetry properties of the Green's function. Boundary conditions of the Green's function: Dirichlet, Robin, Neumann.

14. A. Green's function for Laplace equation on a disk.

    B. Green's function for Laplace equation on a rectangle.

15. Diffusion equation, self-similar solutions, uniqueness, min-max principle.

16. Green's function for the heat equation. Infinite and semi-infinite domains, Neumann and Dirichlet boundary conditions.

17. Green's function for the 1D diffusion equation on a finite domain. Wave equation in 2D and 3D, preliminaries. Uniqueness of solutions.

18. Fourier transform properties. Green's function for the wave equation in nD.

19. Green's function for the wave equation in 1D and 2D.

20. Wave equation. Green's function in 3D. Example: Radiation from an electron on an antenna.

21. Separation of variables in 2D. Laplace, Helmholtz, Schrödinger potentials. Separable coordinate systems in 2D.

22. Sturm-Liouville oscillation theory, Prüfer substitution, comparison theorems.

23. A. Sturm-Liouville theorem. Eigenvalue problem. Properties of zeros, dependence on the eigenvalue. Existence of unique orthogonal basis.

    B. Expanding functions on a Sturm-Liouville eigenbasis. Properties and method.

24. Wave equation on a disk, separation of variables.

25. Wave equation on a circular membrane, Bessel functions, zeros, generating function, integral representations.

26. Wave equation on a circular membrane, expansions using Bessel functions.

27. Solitons, Korteweg-de Vries equations. Advection nonlinear velocity, dispersion.

28. KdV travelling wave solutions, self-similar solutions, inverse scattering method, solitons.

4.13 Ordinary Differential Equations

1. Some Basic Mathematical Models

2. Separable Differential Equations

3. Linear v.s. Non-Linear Differential Equations

4. Linear First Order ODE

5. Applications of Separable Equations
6. Applications
7. Existence and Uniqueness, Linear v.s. Nonlinear
8. Autonomous First Order ODEs, Population Dynamics
9. Exact Differential Equations
10. Euler's Method
11. Review
12. 2nd Order Linear ODEs
13. 2nd Order Linear Homogeneous ODEs with Constant Coefficients
14. Complex Roots of the Characteristic Equation
15. Reduction of Order
16. Nonhomogeneous 2nd Order Linear Equations, Method of Undetermined Coefficients
17. Variation of Parameters
18. Vibrations
19. Higher Order Linear ODEs
20. Forced Vibrations
21. Linear Independence
22. n-th Order Linear ODEs with Constant Coefficients
23. Variation of Parameters for n-th Order Linear ODEs
24. Method of Variation of Parameters for n-th Order Linear ODEs
25. Systems of First-Order Linear Equations
26. On 2×2 Linear Systems with Constant Coefficients
27. Linear 1st Order Homogeneous 2×2 Systerms with Constant Coefficients
28. Linear Systems with Constant Coefficients
29. 3×3 Systems , Fundamental Matrices

30. Repeated Roots

31. Nonhomogeneous First Order Linear Systems

32. Non-linear 2×2 Systems

33. Competing Species

34. Review

35. Analyzing Non-Linear Systems

36. Damped Pendulum with a Stiff String

37. Predator Prey Equations

38. Laplace Transform

39. Using Laplace Transform

40. Laplace Transform of Picewise Defined Functions

41. Review

42. Review